

Yaswanth Deevi

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SUMMARY

A Master's Computer Science student at BTH with a strong interest in machine learning and applied AI. Experienced in building and deploying deep learning models, with hands-on work spanning deepfake detection, anomaly detection, and neural signal processing. Looking for opportunities to apply and grow these skills in an industry or research setting.

EXPERIENCE

Research Intern

Jul 2025 – Sep 2025

Defence Research and Development Organisation (DRDO), Hyderabad

- Proposed and developed an innovative approach for solving the P-Center facility location problem using a combination of evolutionary and deep learning techniques, combining genetic algorithms with DNN surrogates to evaluate potential solutions on large graphs.
- Utilized a pre-trained DNN surrogate to estimate fitness functions, significantly lowering computational costs per generation and accelerating the process by a **factor of 10** compared to conventional evolutionary algorithms.
- Developed sophisticated Gaussian mutations with varying sigmas, depending on population diversity measures, allowing wide exploration initially and fine-grained exploitation closer to convergence.
- Worked alongside senior scientists from DRDO to document our technique, create technical reports, and communicate results, acquiring firsthand knowledge of applying research results in practical engineering advice.

TECHNICAL SKILLS

Programming: Python, C++, SQL, Bash scripting

Deep Learning: PyTorch, TensorFlow, Keras, CNNs, RNNs, Transfer Learning, Model Quantization, Model Pruning, ONNX

Machine Learning: Scikit-learn, XGBoost, LightGBM, Gradient Boosting, Random Forests, Logistic Regression, K-Means, PCA, SHAP

Optimization: Evolutionary Algorithms, Genetic Algorithms, Metaheuristics, Surrogate Modeling, Combinatorial Optimization

Data & Analytics: NumPy, Pandas, SciPy, Matplotlib, Seaborn, Statistical Hypothesis Testing

MLOps & Tools: Git, Docker, Linux, AWS, Jupyter, VS Code

Concepts: Computer Vision, Deepfake Detection, Digital Pre-Distortion, Attrition Modeling, Customer Segmentation, Time-Series Analysis, Flask Deployment

PROJECTS

Neural Digital Pre-Distortion (OpenDPD)

GitHub Repository

- Developed and incorporated two new DPD backbones into the dpdOpen package – MyMLPDPD, a simple MLP-based model with residual connections, and MyLSTMDPD, a BiLSTM network with scaled dot product attention, designed for work with paired IQ signals and adjustable memory.
- Built an indirect learning approach employing sliding windows generation from PA output/input pairs, trained using MSE loss function, gradient clipping, and cosine annealing LR schedule.
- Measured effectiveness of linearization via EVM (%) and NMSE (dB); generated multiple plots of academic format depicting AM/AM and AM/PM responses, PSD, IQ diagrams, and training loss dynamics.
- Performed benchmarking on the DPA.200MHz dataset for both backbones implemented; found that BiLSTM outperformed MLP in compensating for PA nonlinearity due to high memory size.

Deepfake Detection System *(From coursework)*

GitHub Repository

- Optimized EfficientNet-B0 for binary classification of deepfakes in the Celeb-DF dataset; used a sigmoid activation function for output and fine-tuned using binary cross-entropy loss on face crop frames.
- Developed a frame-by-frame inference pipeline involving the use of MTCNN for face detection and alignment; employed a dynamic frame-skipping mechanism (skip every 5 frames) to strike a balance between performance and accuracy.
- Implemented an end-to-end pipeline as a web application written in Flask that ingests videos, adds an FAKE/REAL label to every processed frame along with a confidence score, and encodes the output video as H.264 for playback on browsers.
- Determined final video prediction based on the number of fake or real frames in the video; evaluated system effectiveness using the F1 score, ROC AUC, and confusion matrix for different compression rates and lighting.

Anomaly Detection in Building Heating Systems *(From coursework)*

GitHub Repository

- Created an FP-Growth association rule mining-based streaming anomaly detection pipeline for multi-room building heating system data; adopted a sliding window paradigm with a window size of 6 records, stride of 3, and 50% overlap such that detection delay is less than 30s and avoids tumbling window “missed fault” issue due to boundary

effects.

- Created average-based discretized transaction creation with outside temperature (OO room) as the contextual feature; adopted frequency weighted scoring mechanism with stricter association rule filters (lift, leverage, conviction) to reduce false positives.
- Implemented a pattern stability analysis mechanism that cross-validates mined rules using chronological N fold analysis with respect to each rule's prevalence and confidence drift.
- Compared association rule-based anomaly detection pipeline with 3 unsupervised baselines (Isolation Forest, K-Means, DBSCAN) using identical feature vector sets by creating an agreement matrix, per room disagreement matrix, and a ROC-like curve.

Hybrid Optimization for P-Center Problems *(From research internship)*

GitHub Repository

- Designed surrogate-assisted evolutionary search algorithm using feed-forward DNNs as fitness surrogate, replacing costly distance computations in later generations.
- Designed adaptive Gaussian mutation using sigma scaling exploiting the idea of landscape awareness, helpful in exploration during initial generations and exploitation near convergence.
- Achieved a convergence speedup of 8- to 10-fold in comparison to baseline genetic algorithms, yielding solutions that are within 1.5% of optimal solutions on standard OR-Lib and TSPLIB problems.
- Performed ablative analysis for evaluating the difference between full-exact evaluation, surrogate evaluation, and the combination of both.

HR Analytics and Attrition Modeling *(From coursework)*

GitHub Repository

- Conducted comprehensive end-to-end data analysis using the IBM HR Analytics Dataset with 1,470 entries and 35 attributes; conducted exploratory data analysis on various attributes like demographics, employee satisfaction, tenure, and salary to comprehend the attrition problem.
- Performed thorough statistical hypothesis testing like the Mann-Whitney U test for continuous variables, chi-squared test for categorical variables, and point-biserial correlation for continuous-binary variables; the attributes with maximum importance in attrition were overtime, job satisfaction, and years since last promotion.
- Constructed a logistic regression model (class balanced, $C=1.0$) and random forest classifier using an 80-20 stratified data split; employed logistic regression and random forest classifiers to understand the decision-making process.
- Retention analysis on 8 cohorts of employees hired on a quarterly basis (from Q1 2022 to Q4 2023); retention rate at 3 months, 6 months, 12 months, and 18 months; trends in hiring and leaving each month.

Customer Segmentation and Churn Prediction

GitHub Repository

- Engineered 8 behavioral metrics (tenure, monthly expenditure, NPS, payment lag, logins per month, contract types) on a simulated dataset of 2000 customers, applying Standard Scaler for normalization.
- Applied K-Means clustering with number of clusters $K = 4$ via elbow method and 2-dimensional plotting through PCA to perform customer segmentation into practical segments: Champions, Loyalists, Dormant customers, and At-Risk Customers.
- Built a Gradient Boosting model with 300 estimators, learning rate of 0.05 and max depth of 4 for predicting customer churn, reaching the **CV ROC AUC Score: 0.87 +/- 0.01** through 5-fold stratified cross-validation.
- Generated structured JSON output containing all stages of pipeline execution (customer segments, ROC curve plot, feature importance plot, confusion matrix plot, trend of churners per month).

EDUCATION

M.S. in Computer Science & Engineering

Expected Jul 2027

Blekinge Tekniska Högskola, Sweden

Relevant Coursework: Research Methodologies, Advanced Machine Learning, Deep Learning, Decision Support Systems

B.Tech in Computer Science & Engineering

Dec 2025

Jawaharlal Nehru Technological University Hyderabad

Relevant Coursework: Machine Learning, Data Structures and Algorithms, Computer Networks, Object Oriented Programming, Python Programming.

CERTIFICATIONS

- IIT KGP AI4ICPS – Advanced Machine Learning & Industrial Cyber-Physical Systems
- Smart Interviews – Data Structures, Algorithms & System Design